European Parliament

European space act

This European Parliamentary Research Service paper aims to inform Members on issues related to a forthcoming Commission initiative. It highlights the main choices which may shape the initiative and which Members may wish to explore ahead of formal Commission adoption. Based on documentary and other sources, it reflects the information available at the time of writing.

For further information on this topic, Members and staff of the European Parliament may contact the authors.

ISSUES AT STAKE

- The combined global revenues of Earth observation and positioning-navigation-timing solutions revenues are expected to double between 2023 and 2035, making space services one of the fastest growing economic sectors. According to the European Commission, around 10 % of EU GDP is enabled by satellite signals. However, space congestion hampers further development.
- The probability of a new global 'space race' led the European Council to identify space as a sensitive sector where strategic EU dependencies need to be reduced. A legislative initiative on an EU space law is planned in the European Commission's 2025 work programme (Q2).
- While there is a broad consensus on the relevance of collective EU space assets and services for competitiveness and strategic autonomy, Member States hold different views on the most appropriate legal basis and legislative format for EU action.
- Most stakeholders in the EU are supportive of an EU space act that would address the safety, security and sustainability of space operations and facilitate the creation of a single market for space services, provided it does not hamper the competitiveness of EU space actors by creating additional burdens, and would apply to any entity delivering space services in the EU.
- This situation presents a strategic opportunity for the European Parliament to be proactive and use its tools to shape the proposal (see below), aiming to complement the act with the required financing for space investment in the next EU multiannual financial framework.

Why do we need a European space act?

According to Mario Draghi's report on the future of EU competitiveness, the value of the global space economy reached €572 billion in 2023, and is expected to increase 9 % annually on average until 2035, when it may reach €1 628 billion. The private sector is gaining ground across space sectors: for instance, in 2023, the private SpaceX company took charge of 98 of the 109 launches in the United States. However, no major openness in the space markets has accompanied this development; the space market continues to rely mainly on institutional programmes, which are subject to access restrictions.

The space economy is composed of three main segments: a) the upstream segment, including space asset research, development, manufacturing and launch activities; and b) the downstream segment, entailing all space asset and technology applications. A derived segment, c) has been introduced recently, corresponding to economic activities that result from space activities (such as photovoltaic panels or wireless headsets).

Authors: Clément Evroux, Annastiina Papunen

PE 772.914 – June 2025

The continuity of space services is endangered by **congestion and contestation in the space domain**. According to recent data, the number of operational satellites in orbit around Earth doubled between 2020 and 2023, reaching more than 9 000. Besides those operational satellites, the European Space Agency estimates that more than 12 400 tonnes of space objects are orbiting around Earth. This mass reduces the number of practicable orbits for operational purposes (such as navigation or observation), and increases the risk of collision, which can hamper the provision of space services (the 'Kessler effect'). Geopolitical tensions are also reflected in the space domain, which is increasingly contested. More than 80 countries have already registered at least one in-orbit operational satellite. Whilst not all space-faring countries have the same capacities, their space activities do not necessarily facilitate space exploration or the use of space resources as a global commons. For example, in 2021, Russia conducted an illegal anti-satellite strike.

Preparation of the European space act

Against the backdrop of steady growth in active and inactive objects in orbit and consequent increased risk of collision, as well as the existence of several national space laws in more than a dozen Member States (see below), no level playing field exists across the single market. This hampers EU competitiveness and creates risks for other EU sectoral policies, such as transport and environment. The introduction of common binding obligations on spacecraft manufacturers and space system operators would therefore mitigate this regulatory fragmentation and its detrimental effects on the competitiveness and resilience of EU space assets. Such obligations could focus either on ensuring specific physical properties for spacecraft, or on laying down rules for space traffic management, or both.

European Commission

In 2022, a joint communication by the Commission and the High Representative set the EU's position on space traffic management, establishing rules ensuring the continuity and resilience of EU space services and assets. It also stressed the option of an EU legislative initiative to be published by the end of 2024, expected to address three main challenges:

- **safety**: to tackle increasing space congestion by defining rules on spacecraft and space operation safety, and by improving space situational awareness capabilities;
- **security-resilience**: to strengthen the security of space assets and operations, including cybersecurity requirements in orbit;
- **sustainability**: to reflect the EU commitment to ensuring sustainability, including solutions to reduce the luminosity of spacecraft to preserve astronomical observation from Earth, for instance.

The preparatory work on the EU space law included a public consultation conducted during autumn 2023, which gathered around 40 written responses, mostly supportive of the initiative. However, the proposal could not be finalised by the end of the ninth parliamentary term. The new initiative is therefore the continuation of preparations for a European space act launched during the ninth parliamentary term and included in the Commission's 2024 work programme. Publication was originally planned during the first quarter of 2024, however the Commission informed the Committee on Industry, Research and Energy (ITRE) in April 2024 that publication was postponed.

In September 2024, President Ursula von der Leyen included the preparation of an EU space law in her mission letter to Commissioner-designate Andrius Kubilius, in charge of defence and space. During his hearing in November 2024, Kubilius confirmed that the Commission was considering planning to publish a proposal for a regulation on an EU single market for space in 2025, hopefully during the first half of the year. The initiative is expected to focus on the safety, security and sustainability requirements to create an EU single market for space.

Now Commissioner for Defence and Space, Andrius Kubilius leads the preparation of the file. Henna Virkkunen, Executive Vice-President for Tech Sovereignty, Security and Democracy is

coordinating the preparation work, under her mandate on security and technological sovereignty, and is expected to present the proposal in June 2025.

During his hearing in the European Parliament, Commissioner Kubilius provided his views on complementary aspects of the expected EU space law. This included the role of the legislative initiative in ensuring the sustainability of the space domain, as well as the need to strike an appropriate balance between providing a set of EU rules and the objective of simplification. He also underlined the need to bolster EU strategic autonomy, not least through facilitating the promotion of EU space standards that may facilitate economic opportunities for EU space actors.

The Council and the European Council

In April 2024, the European Council identified space as a sensitive sector where strategic dependencies need to be reduced. The Danish Presidency is expected to initiate discussions on the forthcoming space act in the second half of 2025, in the relevant Council working party.

Member States' positions and interested parties' opinions

Commissioner Kubilius mentioned during his hearing that at least 12 Member States have already adopted a specific regulatory framework for space activities. A screening of the UN repository of space laws, along with the screening of recently adopted national legislation (the 2023 Cyprus space Law) shows a total of 13 relevant pieces of national legislation on space activities in force in EU countries at the end of 2024. Since then, the Italian Parliament is examining a legislative proposal on the space economy and Spain is designing a similar legislative proposal. In Estonia, a space law is expected to enter into force on 1 January 2026.

Sources and scope of national space legislation across the European Union

National regulatory frameworks on space in the EU are based on the United Nations' space treaties. Among these – the Treaty on principles governing the activities of States in the exploration and use of outer space ('the Outer Space Treaty'), including the moon and other celestial bodies, which entered into force in 1967, and the Convention on registration of objects launched into outer space, which entered into force in 1976 – constitute the most important legal sources for such obligations:

- Article VI and Article VII of the Outer Space Treaty provide for the international liability of any States parties to the treaty for damages caused by activities in outer space, and the launch of an object into space from its territory or facility.
- Article II of the Convention mentioned above provides for States parties to the treaty to register each object launched into Earth orbit and beyond in a national registry. Article IV of the same Convention stipulates a set of five sets of information to be provided at registration: a) name of the launching State, b) appropriate designator of the space object or its registration number, c) date and location of launch, d) basic orbital parameters, e) general function of the space object.

Member States' national legislation on space activities tend to follow a similar pattern, including notably three main sets of provisions:

- scope of the obligations, which almost always include launches conducted on national territory (including the maritime domain) and launches conducted by national legal or natural persons;
- procedure applicable to the launch of an object into space, which mostly consists of a licence to organise and carry out the launch, to be agreed by a central authority upon compliance with economic and environmental conditions;
- organisation of the national register of space objects.

Member States have deliberated on the background, challenges and policy options regarding the EU approach to mitigating the risks of an increasingly congested and contested space domain in the space segment of the Competitiveness Council. With a set of conclusions adopted in 2022 on an EU

approach to space traffic management (STM), whilst welcoming the definition itself, Member States invited the Commission to facilitate coordination between Member States to address STM legislation, to foster the convergence of national positions on an EU STM approach. The conclusions also caution the Commission to consider the risks and benefits of developing EU STM legislation on the competitiveness of EU industry. With a set of conclusions adopted in 2023, the Council reminds the Commission of the need to comply with Member States' competences and prerogatives, including in safeguarding their national security. In 2024, the Ministers in charge of space held a discussion on the objectives of an EU legislative initiative (safety, resilience and sustainability of the space activities in the EU), and on the possible risks of the development of foreign-based standards and rules on STM.

The deliberations of the Competitiveness Council have not yet paved the way to full convergence among Member States' views regarding the opportunity and role for such a legislative proposal.

In March 2025, a non-paper by Finland and Sweden expressed their concern regarding the existence of an appropriate legal basis for an EU legislative initiative on space. The letter of Article 4(3) and of Article 189(2) of the Treaty on the Functioning of the European Union (TFEU) seem to exclude any EU initiative that would impede Member States from exerting their own competence on space, or any harmonisation of the laws and regulations of the Member States in the field of space policy. The two Member States deem that the dual-use nature of most space technologies should prevent the EU from intervening in such a way as to create any impediment to national competence. However, they would be open to reinforcing space safety, security and sustainability through EU non-legislative proposals, such as the establishment of labels. Compliance with non-legislative initiatives could be made mandatory to access EU procurement opportunities.

In a joint non-paper of December 2024 (not published online), a group of three Member States (Germany, Italy, Slovakia) support the legislative initiative, but spell out requirements to ensure the proposal complies with the proportionality principle, which might also facilitate some convergence with the position of Finland and Sweden. The three Member States consider that, while the proposal should aim at ensuring a robust EU space market, with a 'common level playing field' for EU and non-EU actors, its scope should focus on 'commercial space activities, [and exclude] sovereign space applications'. They infer from this expected scope the possibility to use Article 114 TFEU (harmonisation of law) as a possible legal basis. On safety, the three Member States agree that the legislative proposal should include rules on space traffic management, which should be designed in an inclusive way, to facilitate further proceedings at a multilateral level. The non-paper invites the Commission to define only the essential requirements of the standards necessary to implement the proposal. Such standards should then be developed by the industrial and national authority representatives in the appropriate standardisation bodies. On security, these Member States would favour extending to space activities the scope of the Directive on the resilience of critical entities ('CER Directive') and the Directive on measures for a high common level of cybersecurity across the Union ('NIS2 Directive').

Other Member States, such as France, broadly support a legislative proposal at EU level, providing compliance with the principle of subsidiarity is ensured. The two chambers of the French Parliament have adopted resolutions supportive of an EU space law. In 2023, the National Assembly adopted a resolution on adopting an EU space law, inviting the EU to clarify the internal market rules for space activities and to ensure a common level playing field between EU and non-EU businesses. In 2024, the French Senate adopted a resolution on EU regulation of space traffic management, where it stressed that the national legislations create a risk of fragmentation of the internal market, which should be addressed by an EU legislative framework that should aim at avoiding creating disproportionate administrative burden for space operators active on the EU market. On a similar note, in January 2025, the Danish minister for higher education and science delivered a speech at the 17th European Space Conference, underlining the positive role of fostering a competitive EU space market to handle EU security.

The European space industry has also conveyed its views and expectations on the expected proposal. This is notably the case of ASD-Eurospace (Eurospace), which published a comprehensive position paper in late 2023. In general, Eurospace welcomes an EU space act that could spur the finalisation of an EU internal space market, improve capacity to attract investment with a clearer legal framework and ensure a level playing field with non-EU competitors. These positive expectations are accompanied by advice. As for the methodology, Eurospace calls on the Commission to prepare a thorough impact assessment, to involve the industry in implementing the new legislation, and to provide sufficient time to organise a smooth transition. On the substance, Eurospace stresses the importance of the geographical scope of the proposal. While supporting the inclusion of all entities delivering space-based services in the EU, regardless of their jurisdiction, Eurospace stresses the need to retain a flexible approach regarding the delivery of space-based services by EU operators in third countries. In terms of material scope, Eurospace also invites the Commission and the co-legislators to weigh whether the necessary recognition of the entire space sector is covered in the NIS2 Directive. Eurospace stresses the opportunity to harness the proposal to lay down the first legal definition for space resilience at international level. It also proposes an EU certification or label, to be granted to compliant EU space actors. More generally, Eurospace thinks that the EU should promote its standards, when possible. It also calls on the EU to harmonise the liability schemes applying to space operations across the EU. On security, it calls for all new satellites to include on-board cybersecurity. On safety, it stresses the need to reinforce EU space situational awareness capabilities, noting the EU's dependency on the United States. On sustainability, it also calls for major EU investment in research and development to allow for the deployment of technological solutions, for instance in the field of active debris removal. Some recommendations from Eurospace align with other stakeholders' views. For instance, the need to avoid unnecessary regulatory burden, and to ensure cybersecurity by default in space systems is compatible with the views conveyed by SME4space, an umbrella organisation of European SMEs active in the space sector, which issued a paper in 2024 on the future of EU space policy.

Other economic stakeholders have also provided their views. In Italy, Confindustria, the main association representing manufacturing and service companies in Italy, expressed its preference for the harmonisation of the liability schemes in March 2025. In Germany, in late 2024, following the presentation of an outline for a federal space law, the German aerospace industries association avowed its preference for aligning this with EU legislation.

Space stakeholders from third countries are also following preparation of the proposal closely. For instance, the US think-tank Centre for Space Policy and Strategy published an article in 2024, in which it stresses the need for US industry and authorities to mitigate the risks incurred by US space companies. While the compliance costs of EU legislation might hinder the competitiveness of US space systems and services, the price of non-compliance might be even higher should the EU succeed in setting space standards with global reach. United Kingdom space stakeholders may face similar dilemmas, with more intensity due to the UK's membership of the European Space Agency (ESA). A 2024 article from the Chambers and Partners research firm estimates that at least one third of UK space sector annual revenue depends on exports.

European Parliament

According to Parliament's Rules of Procedure (Annex 6), EU space policy falls under the remit of the ITRE committee.

In September 2022, the ITRE committee asked the Commission a question for oral answer on an 'EU Approach for Space Traffic Management – An EU contribution addressing a global challenge'. In October 2023, while the Commission public consultation was ongoing, Parliament's Panel for the Future of Science and Technology (STOA) organised an event entitled 'Future of space - the sustainable path', which discussed many elements of the expected space law.

In November 2023, Parliament welcomed the space law initiative and the aim to establish 'a common, harmonised and coherent security, safety, and sustainability framework' in its resolution on the 'Strategic Compass and EU space-based defence capabilities'. In Parliament's view, the law would improve the resilience of EU space services, strengthen EU strategic autonomy, tackle market fragmentation and enhance EU space companies' competitiveness. Notably, Parliament asked for 'significant emphasis' on information security and protective measures and to ensure access to space in crises. It was suggested that private space companies could be considered 'as private entities with a public service mission' to ensure continuity of service. Furthermore, Parliament asked the Commission to include registration, liability and environmental standards in the forthcoming legislation. During then Commissioner-designate Kubilius's 2024 hearing, the initiative to propose an EU space law was broadly supported across the political groups.

In September 2024, the ITRE committee discussed the Commission's implementation report on the EU space programme and gave a favourable opinion. In May 2025, ITRE organised a hearing on 'Reinventing European space governance and boosting industry: responding to challenges', and exchanged views with the Director-General of ESA, Josef Aschbacher. Aschbacher declared he had not been consulted on the proposal. Also in May 2025, Parliament adopted a resolution on a revamped long-term EU budget, which stresses that a secure space sector is critical to EU sovereignty and requires sustained investment, highlighting the link between the proposed EU space act and the continuity of EU space assets and services.

FURTHER READING

Pataki G.Z., EU capabilities in space: Scenarios for space security by 2050, EPRS, European Parliament, March 2025.

Evroux C., EU space policy: state of play, EPRS, European Parliament, November 2024.

DISCLAIMER AND COPYRIGHT

This document is prepared for, and addressed to, the Members and staff of the European Parliament as background material to assist them in their parliamentary work. The content of the document is the sole responsibility of its author(s) and any opinions expressed herein should not be taken to represent an official position of the Parliament.

Reproduction and translation for non-commercial purposes are authorised, provided the source is acknowledged and the European Parliament is given prior notice and sent a copy.

© European Union, 2025.

eprs@ep.europa.eu (contact)

https://eprs.in.ep.europa.eu (intranet)

www.europarl.europa.eu/thinktank (internet)

http://epthinktank.eu (blog)